

Journal of Fluid Mechanics
Attn. Professor Sutanu Sarkar, Associate Editor

Re: revised submission developed from JFM-2026-1781

Dear Professor Sarkar,

We submit for consideration a revised manuscript, developed from JFM-2026-1781, which you handled as associate editor and which was declined in August 2026. The decision letter noted that a subsequent submission containing original material from that manuscript should cite its number and be assigned initially to the associate editor who handled it; we therefore cite JFM-2026-1781 and, as you were that editor, bring the revised work back to you directly.

The three reports converged on two points. First, the central spectral prediction had been tested only against the model's own kinematics, so the agreement carried little weight. Second, the paper was long and did not separate its new results from established ones. We have rebuilt the manuscript around both.

The model is now measured against an independent benchmark. A 128^3 strained-box direct numerical simulation, together with exact rapid-distortion companions reduced to the same one-dimensional observables, provides the spectral comparison (§4.6, figure 11), and the departure from linear theory is reported as a function of strain rapidity (figure 12). The linear reference is now the exact rapid-distortion solution projected onto the line, which is broadband rather than a rigid spectral translation (§4.2, figure 8); the rigid-translation object of the earlier version is identified as the model's own kinematics. This addresses the benchmark concern raised by all three referees and Referee 3's objection to the linear reference.

Section 5 no longer offers a closed single-line functional. It states the obstruction exactly, derives sharp bounds on the rapid pressure-strain from line data alone, measures the axisymmetry assumption rather than imposing it, and reports the component-ordering deficiency against the direct simulation, localising it in the closure. Where the single-line representation falls short, the shortfall is now quantified.

The leading-edge-noise application is carried through (§7): the computed distorted inflow spectrum is supplied to Amiet's theory and the predicted level compared with four published experiments (figures 17–21).

The manuscript has been reorganised and shortened. Implementation detail and the catalogue of intervention variants are moved to the supplementary material, established and new results are separated, the notation is fixed once, and the introduction has been revised to include the near-edge-distortion literature the referees identified.

A point-by-point response to each referee accompanies this letter. We are grateful to the referees; their reports directed the revision toward the validation and the scope the work needed.

Yours sincerely,
Sparsh Sharma and Alan R. Kerstein